

Literature Review

Table 1. Summary of Recent Related Work on Graph-Based sEMG Analysis and Gesture Recognition (2022–2025)

Title	Author& Year	Dataset	Application area	Method / model	Metrics	Strengths	Limitations	Research gap	Relation to this work
Improved Multi-Stream Convolutional Block Attention Module for sEMG-Based Gesture Recognition	Wang et al., 2022	NinaPro DB1; custom sEMG–ACC	Prosthetic and HMI control	Multi-stream 1D-CNN + GRU + CBAM attention	DB1 Acc: 89.70% ; Custom Acc: 94.10% ; delay <300 ms	All 52 DB1 gestures; temporal and channel features	No electrode graph; unclear cross-subject test	Graph-based electrode relationships	Direct non-GNN DB1 baseline
A Hybrid Model Based on ResNet and GCN for sEMG-Based Gesture Recognition	Xu & Jiang, 2023	NinaPro DB1	Gesture recognition	ResNet–GCN; muscle-synergy graph; joint voting	Acc: 90.07%	GCN on DB1; muscle-synergy learning	Graph edges unclear; no attention	Clear adaptive graph and strict subject split	Main GCN baseline
Stretchable Array Electromyography Sensor With Graph Neural Network for Static and Dynamic Gestures Recognition System	Lee et al., 2023	Custom 8-channel sEMG; 18 gestures	Wearable and prosthetic control	Spatiotemporal GNN; electrodes as nodes; self-attention adjacency	Avg. Acc: ≈97% ; long-term Acc: 96.61%	Adaptive input-specific graph; 250 ms windows	Custom sensor; small dataset	Public-dataset and unseen-subject validation	Reference for adaptive graph attention
A Spatio-Temporal Graph Convolutional Network for Gesture Recognition From High-Density Electromyography	Zhong et al., 2023	Public HD-sEMG; 65 gestures	Prosthetic and HMI control	STGCN-GR; functional-connectivity electrode graph + temporal convolution	Acc: 91.07%	Spatial and temporal learning; 65 gestures	High-density EMG differs from DB1	Adapt graph to sparse 10-channel EMG	Reference for spatiotemporal GCN

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A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals	Vipulanandan et al., 2024	Custom Myo data; 8 electrodes, 8 subjects	Real-time prosthetic and AR control	Real-time GNN; muscle-activation graph	Acc: 99% ; runtime: 48 ms	High accuracy; fast inference	Small custom dataset; graph details limited	Larger public and cross-subject testing	Real-time performance reference
Hand Gesture Recognition Using sEMG Signals With a Multi-Stream Time-Varying Feature Enhancement Approach	Shin et al., 2024	NinaPro DB1 and DB9	Prosthetic and HMI control	Multi-stream TCN/CNN/LSTM + channel attention	DB1: Acc 94.31% , Precision 95.60% , Recall 94.31% , F1 94.08%	Strong DB1 result; multiple metrics	No graph; 70/15/15 split	Graph model under subject-based evaluation	Strongest non-GNN DB1 benchmark
Dynamic Graph Transformer-Convolutional Network for Multi-Channel EMG Gesture Recognition	Wang et al., 2025	Public multichannel EMG dataset	Biomedical gesture recognition	Dynamic GCN–Transformer; time-varying graph + self-attention	Acc: 97.24%	Dynamic channel relations; long-range temporal learning	Dataset and protocol unclear; not DB1	Reproducible evaluation on NinaPro	Future dynamic graph architecture

Problem Statement

Existing sEMG-based studies mainly focus on classifying discrete hand gestures, although practical prosthetic control requires estimation of continuous finger and hand positions. Graph-based sEMG research also commonly uses custom or high-density electrode datasets, while continuous pose regression using low-channel public datasets remains less explored.

Another major problem is inter-subject variability. The same movement can generate different sEMG patterns across individuals. Models evaluated using random or within-subject splits may therefore perform poorly when applied to a completely unseen person.

This study addresses these problems by formulating hand-pose estimation as **a** cross-subject multi-output regression task. Windowed 10-channel sEMG signals from NinaPro DB1 will be used to predict synchronized 22-channel glove measurements through a graph-based regression model, with test subjects kept completely unseen during training.

Research Objective

Main Objective

To develop and evaluate a graph-based multi-output regression model for continuous hand-pose prediction from surface EMG signals and assess its generalization to previously unseen subjects using NinaPro DB1.

Specific Objectives

1. Prepare synchronized EMG–glove windows from NinaPro DB1.
2. Use 10-channel windowed EMG as input and 22-channel glove measurements as regression targets.
3. Develop conventional and deep-learning regression baselines.
4. Construct a graph representing relationships among the EMG channels.
5. Develop a GCN- or GAT-based multi-output regression model.
6. Apply subject-independent training, validation, and testing splits.
7. Compare graph-based regression with conventional regression and 1D-CNN baselines.
8. Evaluate performance using MAE, RMSE, R^2 , and Pearson correlation.
9. Analyze overall, per-subject, and per-glove-channel prediction performance.